

Functional Test Cases: Entity Management (CRUD Operations)

Library Management System

Test Design Team (Non-Coders/Specification Experts)

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Note: These test cases are designed from a black-box, user-centric perspective based on the SRS document. Actual results and status are to be filled during test execution. Test case design incorporates techniques like Equivalence Partitioning and Boundary Value Analysis.

I.B Functional Test Cases: Entity Management (CRUD)

Table 1: Entity Management Test Cases

Test Case ID	Test Title	Test Steps	Expected Results
Sub-Section: Book Management (SRS 3.3)			
TC_BOOK_CREATE_001	Successful new book creation by Librarian	- Login as Librarian. - Navigate to 'Book Management'. - Click 'New Book'. - Fill all required fields (Name, ISBN) with valid, unique data. Select existing Author and Genre. - Submit the form.	- System accepts the data. - A success message is displayed. - The new book appears in the book list.
TC_BOOK_CREATE_002	Attempt to create a book with missing required fields	- Login as Librarian. - Navigate to 'Book Management'. - Click 'New Book'. - Leave the 'Name' or 'ISBN' field blank. - Submit the form.	- System rejects the creation. - An error message indicating required fields is displayed. - The book is not created.
TC_BOOK_CREATE_003	(RBAC) Verify Member cannot create a book	- Login as a Member. - Navigate to 'Book Management'.	- The 'New Book' button/option is not visible or is disabled. - Direct access to the create book API endpoint (e.g., POST /api/book/add) should be denied.
TC_BOOK_READ_001	Verify all users can view the list of books	- Login as Librarian. Navigate to 'Book Management'. - Logout. Login as Member. Navigate to 'Book Management'.	Both Librarian and Member can successfully view the list of all books.
TC_BOOK_UPDATE_001	Successful book update by Librarian	- Login as Librarian. - Navigate to 'Book Management'. - Select a book and choose to edit it. - Modify a field (e.g., the summary). - Save the changes.	- System accepts the changes. - A success message is displayed. - The book list/details reflect the updated information.
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Table 1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC.BOOK.DELETE_001	Successful book deletion by Librarian	• Pre-condition: Create a book that has no associated borrowals. • Login as Librarian. • Navigate to 'Book Management'. • Select the book and click 'Delete'. • Confirm the deletion.	• System accepts the deletion. • A success message is displayed. • The book is removed from the book list.
TC.BOOK.DELETE_002	(Referential Integrity) Attempt to delete a book that is currently borrowed	• Pre-condition: A book exists and is part of an active borrowal record. • Login as Librarian. • Navigate to 'Book Management'. • Attempt to delete the borrowed book.	• System prevents the deletion. • An appropriate error message is displayed (e.g., "Cannot delete a book with active borrowals"). • The book is not deleted.
Sub-Section: Author Management (SRS 3.4)			
TC.AUTHOR.CREATE_001	Successful new author creation by Librarian	• Login as Librarian. • Navigate to 'Author Management'. • Click 'New Author'. • Fill the required 'Name' field with valid data. • Submit the form.	• A success message is displayed. • The new author appears in the authors list.
TC.AUTHOR.DELETE_001	(Referential Integrity) Attempt to delete an author linked to a book	• Pre-condition: An author is assigned to at least one book in the system. • Login as Librarian. • Navigate to 'Author Management'. • Attempt to delete the author.	• System prevents the deletion. • An error message is displayed (e.g., "Cannot delete author assigned to existing books"). • The author is not deleted.
Sub-Section: Genre Management (SRS 3.5)			
TC.GENRE.CREATE_001	Successful new genre creation by Librarian	• Login as Librarian. • Navigate to 'Genre Management'. • Click 'New Genre'. • Fill the required 'Name' and 'Description' fields with valid data. • Submit the form.	• A success message is displayed. • The new genre appears in the genres list.
TC.GENRE.DELETE_001	(Referential Integrity) Attempt to delete a genre linked to a book	• Pre-condition: A genre is assigned to at least one book in the system. • Login as Librarian. • Navigate to 'Genre Management'. • Attempt to delete the genre.	• System prevents the deletion. • An error message is displayed (e.g., "Cannot delete genre assigned to existing books"). • The genre is not deleted.
Sub-Section: User Management (SRS 3.7)			
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Test Case ID	Test Title	Test Steps	Expected Results
TC_USER.CREATE_001	Successful new user (Member) creation by Librarian	• Login as Librarian. • Navigate to 'User Management'. • Click 'New User'. • Fill required fields (Name, Email, Password) with valid, unique data. Set 'isAdmin' flag to false. • Submit the form.	• A success message is displayed. • The new user appears in the user list.
TC_USER.CREATE_002	Attempt to create a user with an existing email	• Login as Librarian. • Navigate to 'User Management'. • Attempt to create a new user with an email address that is already in use.	• System rejects the creation. • An error message like "User already exists" is displayed.
TC_USER.READ_001	(RBAC) Verify Member cannot access User Management	• Login as a Member.	• 'User Management' link/feature is not visible or is disabled. • Direct navigation to the user management URL is denied.
TC_USER.DELETE_001	(Referential Integrity) Attempt to delete a member with active borrowals	• Pre-condition: A member has one or more active (not returned) borrowal records. • Login as Librarian. • Navigate to 'User Management'. • Attempt to delete the member.	• System prevents the deletion. • An error message is displayed (e.g., "Cannot delete member with active borrowals"). • The user is not deleted.
Sub-Section: Borrowal Management (SRS 3.6)			
TC_BORROW.CREATE_001	Successful borrowal creation by Member	• Pre-condition: A book is available (isAvailable=true). • Login as Member. • Navigate to 'Borrowal Management'. • Click 'New Borrowal' and select an available book. • Submit the form.	• A new borrowal record is created with 'Borrowed' status. • A success message is displayed. • The book's availability status is updated to 'false'.
TC_BORROW.CREATE_002	Attempt to borrow an unavailable book	• Pre-condition: A book is unavailable (isAvailable=false). • Login as Member. • Attempt to create a borrowal for the unavailable book.	• The system prevents the borrowal. • An appropriate error message is displayed (e.g., "Book is not available").
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Table 1 – continued from previous page

Test Case ID	Test Title	Test Steps	Expected Results
TC_BORROW_READ_001	Verify Member can only see their own borrowal history	• Pre-condition: Multiple members have borrowal records. • Login as Member 1. • Navigate to 'Borrowal Management'.	• The list only displays borrowals where the memberId matches Member 1's ID. • Borrowals for other members are not visible.
TC_BORROW_UPDATE_001	Successful borrowal update by Librarian (mark as returned)	• Pre-condition: An active borrowal exists. • Login as Librarian. • Navigate to 'Borrowal Management'. • Select the active borrowal and edit it. • Change the status to 'Returned'. • Save the changes.	• The borrowal record status is updated. • The associated book's 'isAvailable' status should ideally be updated to 'true'. • A success message is displayed.
Sub-Section: Review Management (SRS 3.8)			
TC_REVIEW_CREATE_001	(Assumed) Successful review creation by Member	• Pre-condition: Based on assumed member capabilities if UI were present. • Login as Member. • Navigate to a specific book's detail page. • Submit a new review with a star rating and text.	• A success message is displayed. • The new review is visible on the book's page.
TC_REVIEW_CREATE_002	Attempt to create a review with invalid data	• Login as Member. • Attempt to submit a review with a required field missing (e.g., star rating) or an invalid value (e.g., 6 stars).	• System rejects the submission. • An appropriate error message is displayed.
TC_REVIEW_DELETE_001	Successful review deletion by Librarian	• Pre-condition: A review exists for a book. • Login as Librarian. • Navigate to the book or a review management section. • Select a review and delete it.	• The review is successfully deleted. • The review is no longer visible.